

Capture The Flag

Mission: Learn how Angular applications are built using reusable components.

Objective: Transform your first Angular application into a modular application using reusable components while following a basic Git workflow.

Flag 0 – Return to Base

Mission

Open the Angular project you created during the previous activity.

Tasks

- Open **Visual Studio Code**.
- Open your Angular project folder.
- Open the integrated terminal.
- Verify that the project still runs.
- ng serve

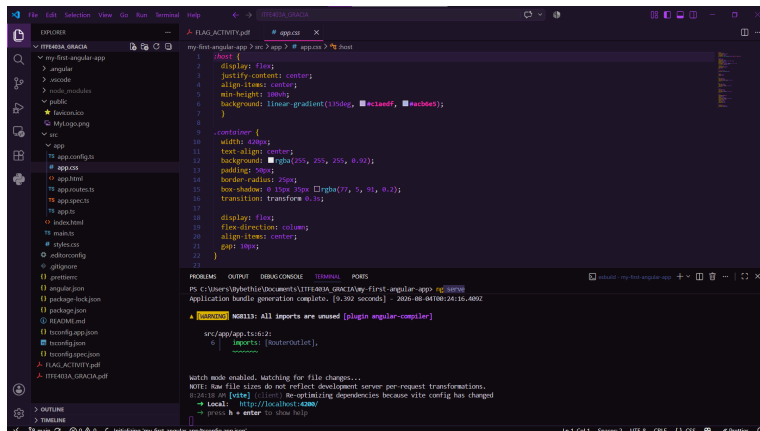
Open:

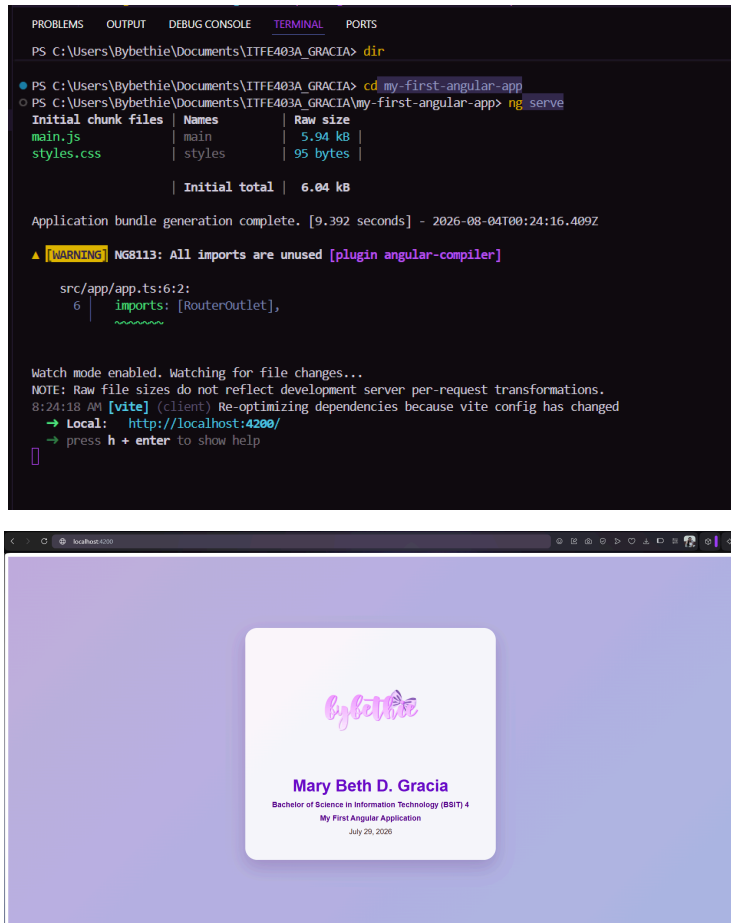
http://localhost:4200

If your Angular welcome page appears, proceed.

Evidence

- VS Code with project opened
- Terminal showing `ng serve`
- Browser displaying the Angular application





Flag 1 – Mission Branch

Mission

Every new feature must have its own Git branch.

Create a new branch named:

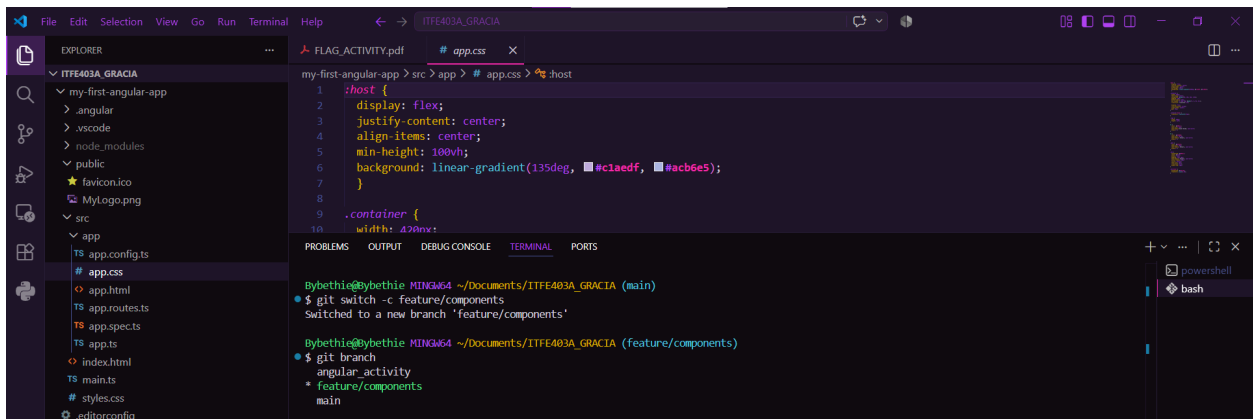
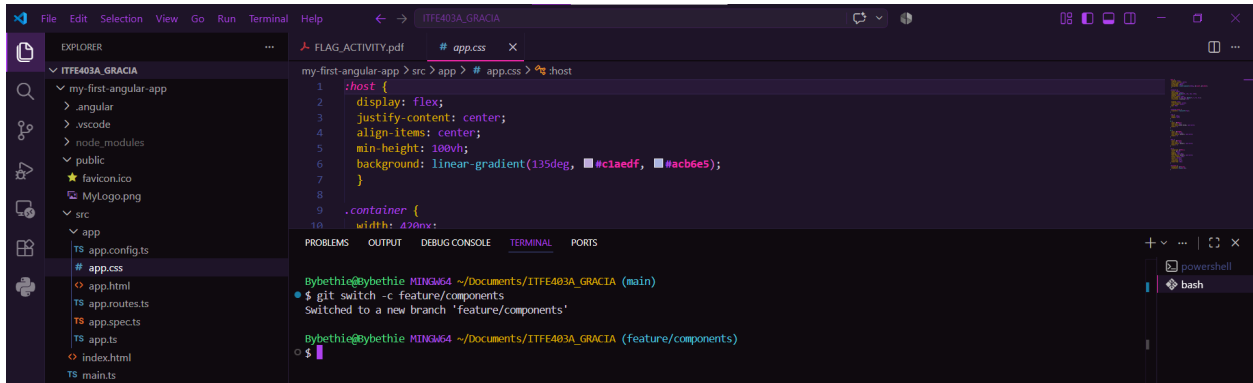
feature/components

Commands:



Evidence

- Terminal showing the successful branch creation
- Output of `git branch`



Flag 2 – Explore the Base

Mission

Investigate your Angular project before making changes.

Locate the following:

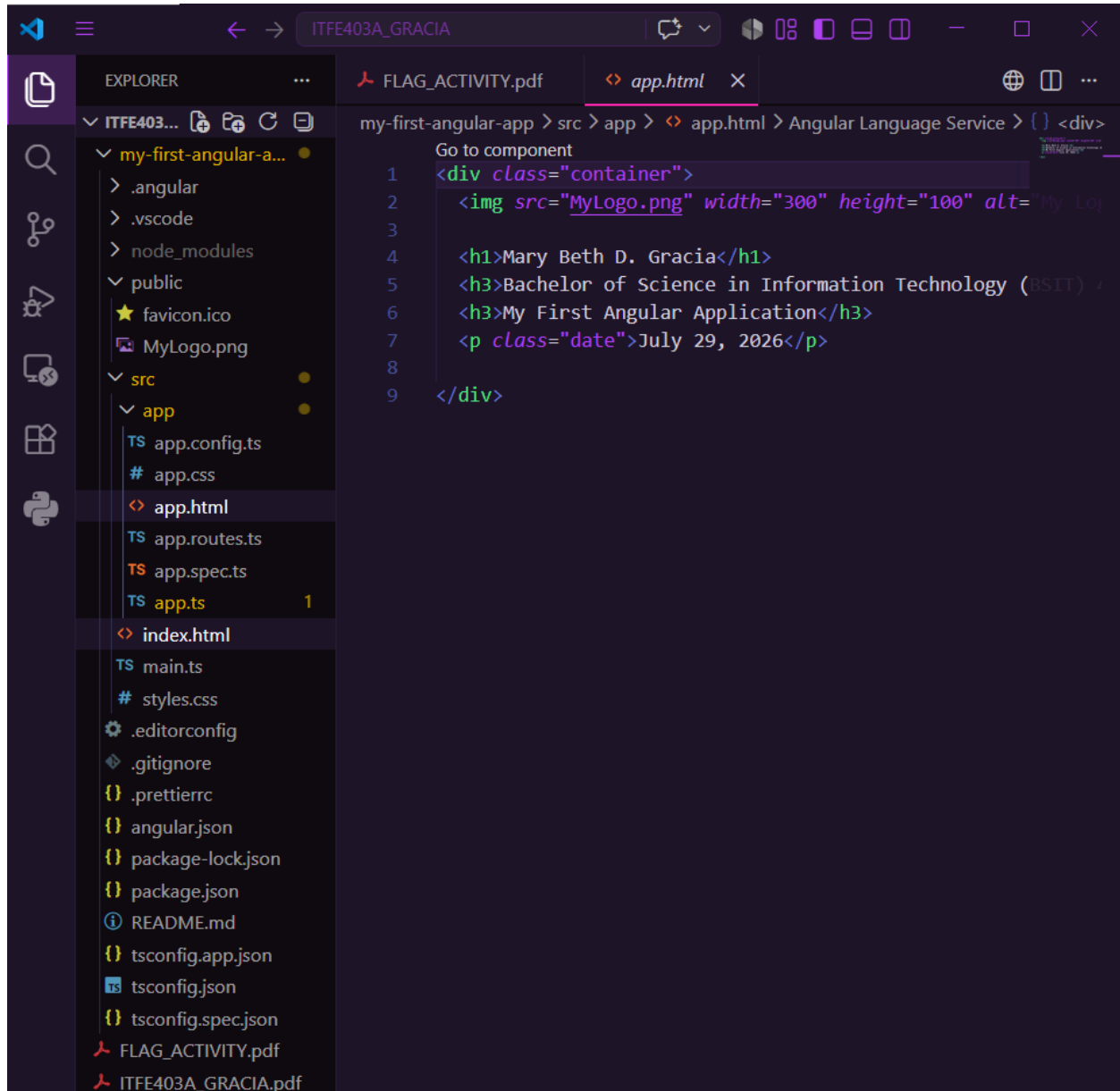
- app.html
- app.ts
- app.css
- app.spec.ts

Understand that these files belong to the **App Component**, the root component of the application.



Evidence

- Explorer showing the files
- app.html opened



Flag 3 – Clear the Battlefield

Mission

Remove the default Angular welcome page.

Delete everything inside:

app.component.html

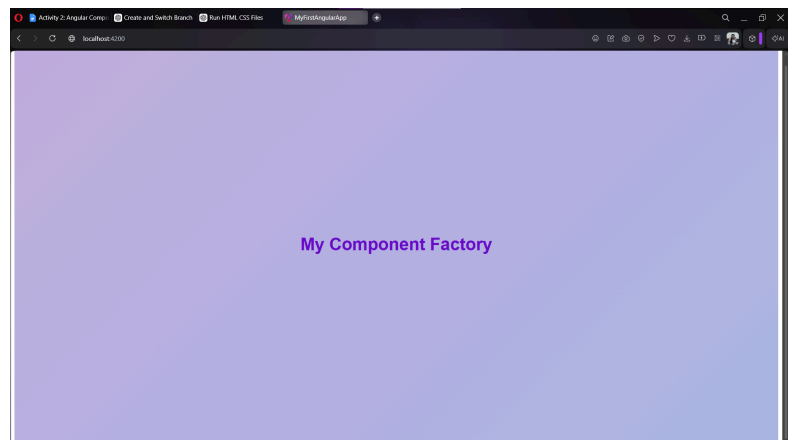
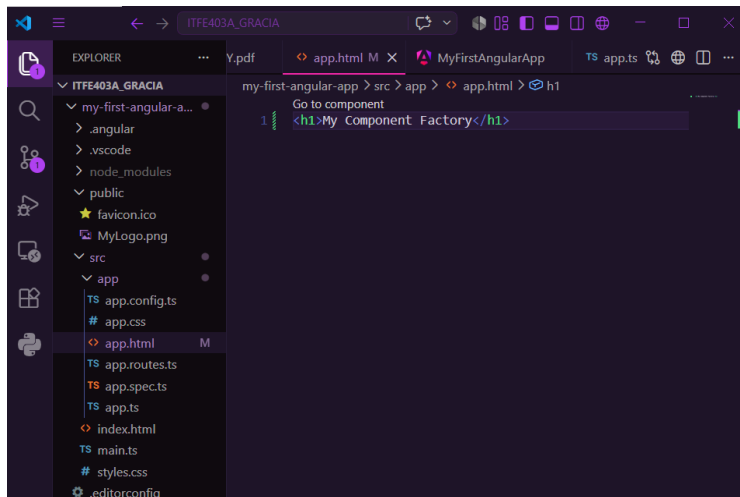
Replace it with:

```
<h1>My Component Factory</h1>
```

Verify the browser updates.

Evidence

- Browser showing only the heading
- HTML file



Flag 4 – Build the Headquarters

Mission

Create a reusable Header component.

Generate:

ng generate component components/header

Angular should automatically generate:

header.ts

header.html

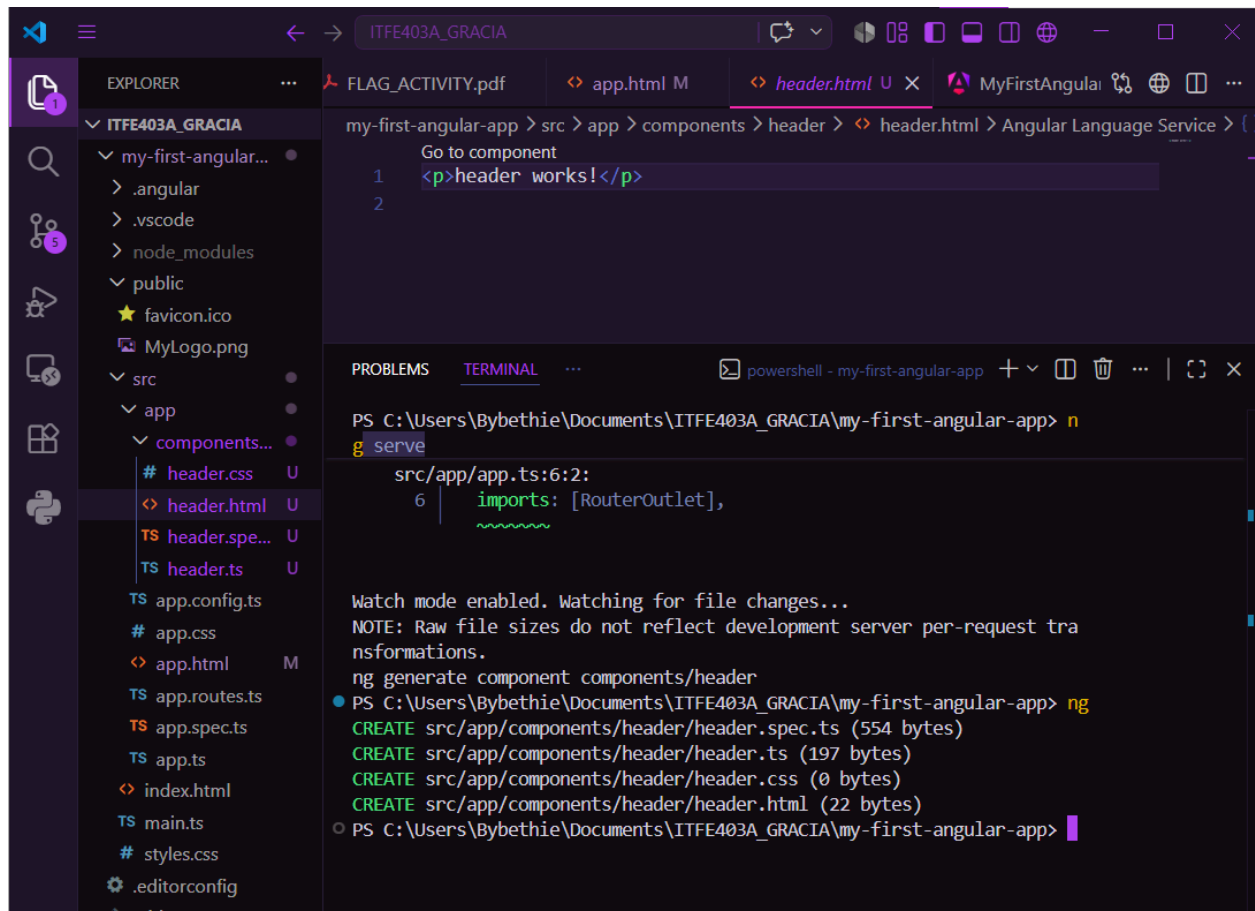
header.css

header.spec.ts



Evidence

- Terminal
- Explorer showing generated files



Flag 5 – Raise the Banner

Mission

Inside the Header component, create a simple header.

Example:

```
<header>
  <h2>Don Bosco Student Portal</h2>
</header>
```

Use the component inside:

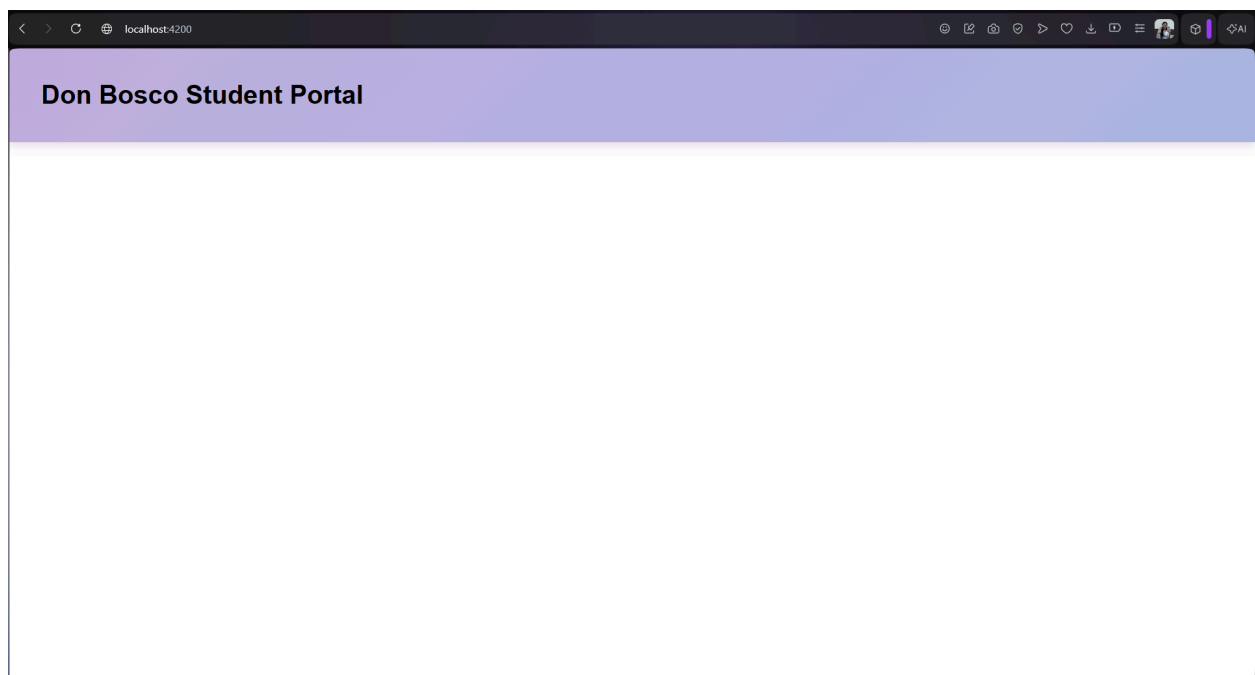
```
app.html
<app-header></app-header>
```

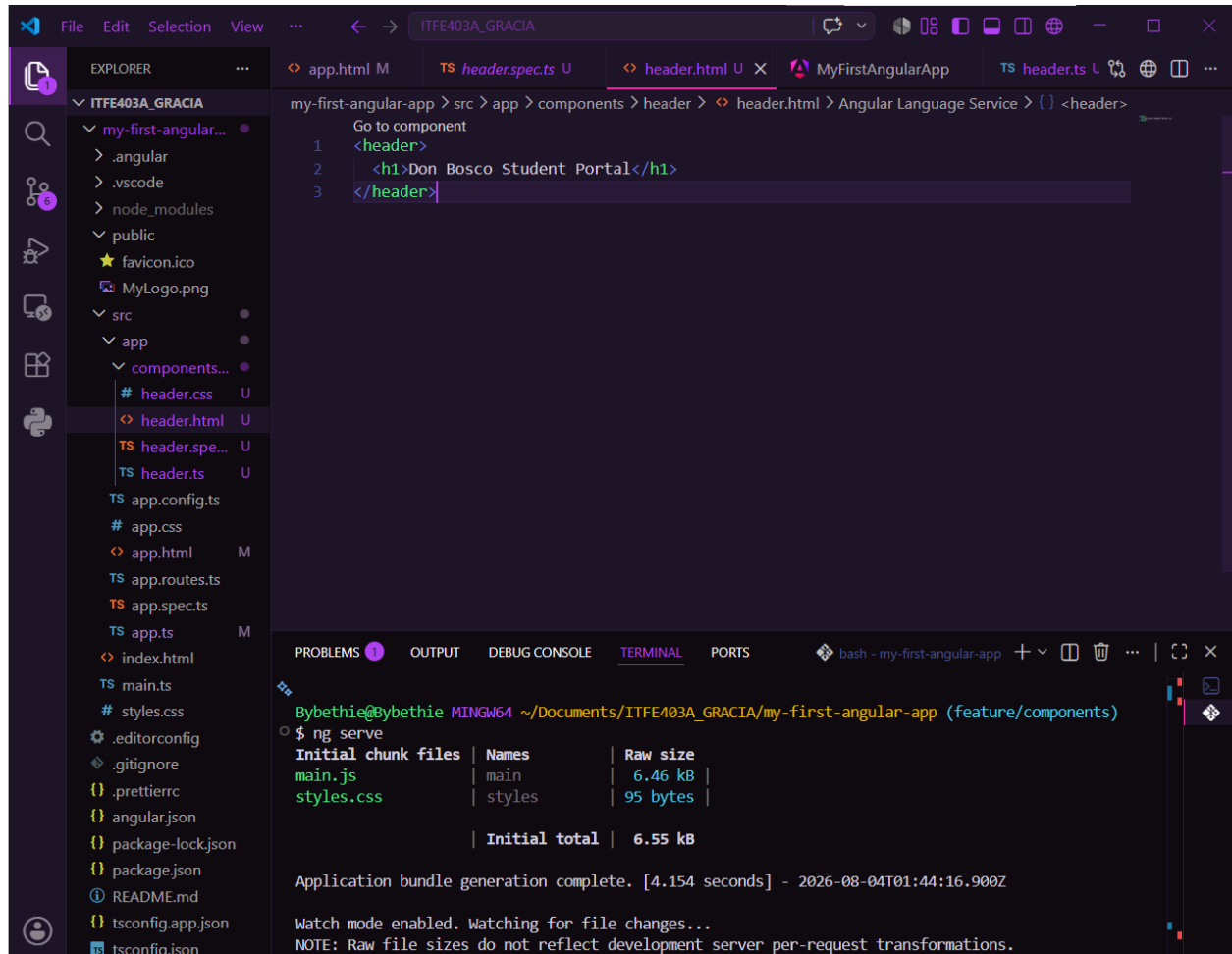
Verify it appears in the browser.



Evidence

- Browser
- HTML code





Flag 6 – Secure the Exit

Mission

Create a Footer component.

Use the same command previously, but now for the footer.

Display:

```
<footer>
```

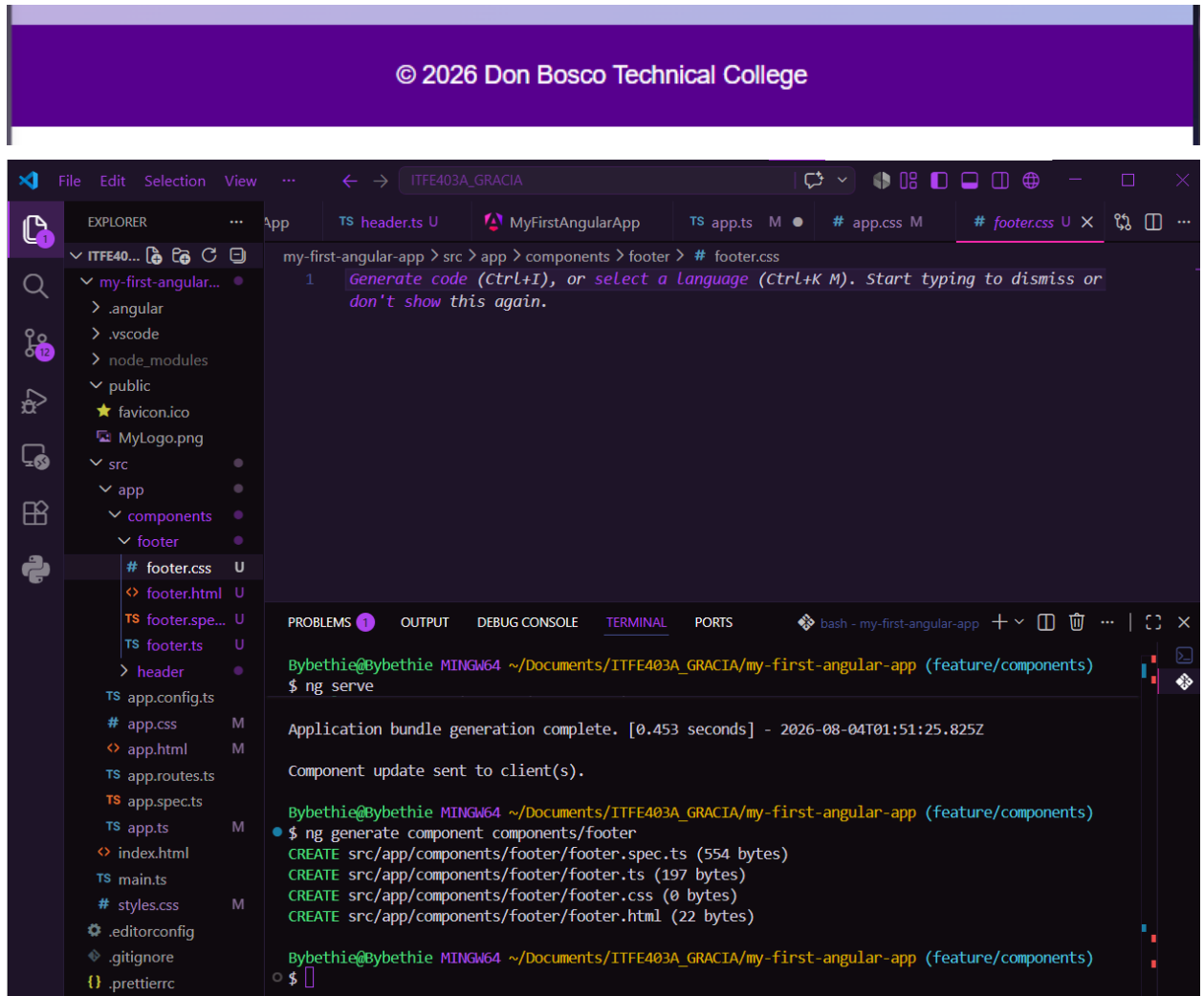
© 2026 Don Bosco Technical College

</footer>

Insert it into the App Component.

Evidence

- Browser
- Generated files



Flag 7 – Create the Profile Card

Mission

Generate a reusable Profile Card.

Use the command to create a new component called profile-card.

Create a card containing:

- Profile Picture
- Name
- Course
- Year Level

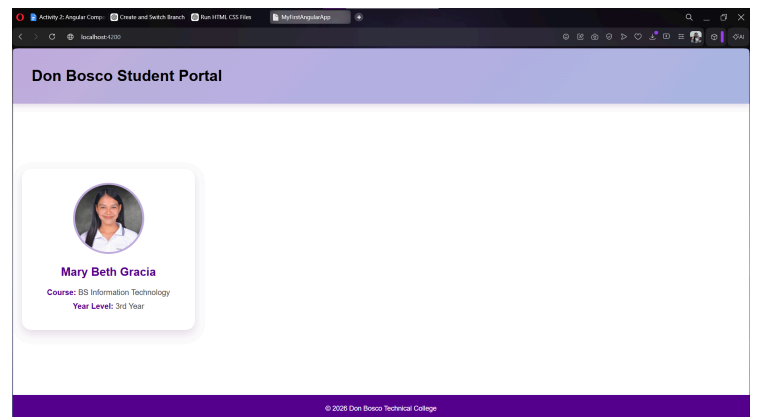
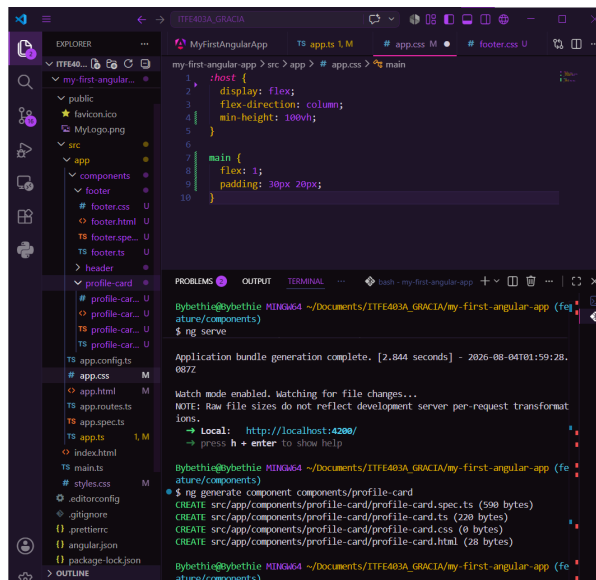
(Design is optional)

Insert it into:

app.html

Evidence

- Browser
- Profile Card component



Flag 8 – Produce Announcement Cards

Mission

Generate one reusable Announcement Card.

Use the same command, creating the component announcement-card.

Design a card containing:

- Announcement Title
- Description
- Date

Do **NOT** create three separate components.

Instead, reuse the same component.

Example:

```
<app-announcement-card></app-announcement-card>
```

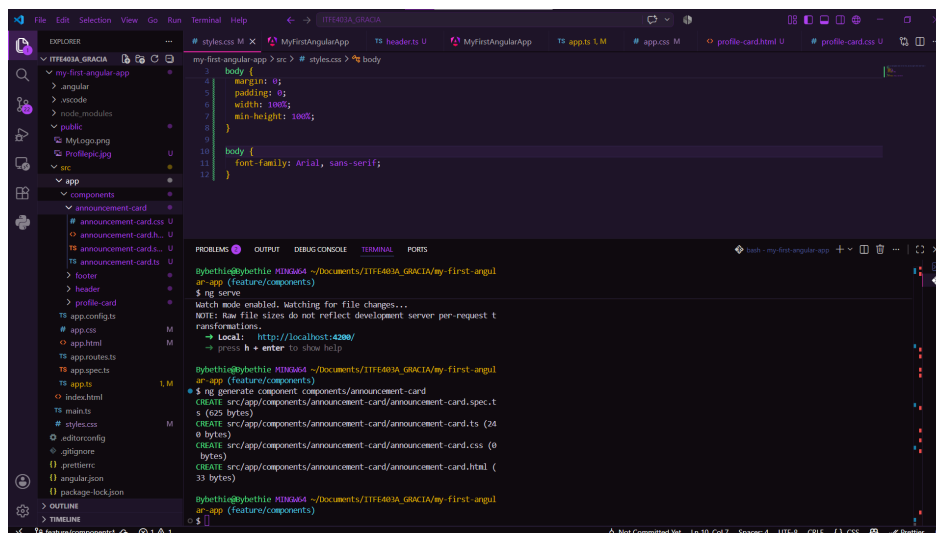
```
<app-announcement-card></app-announcement-card>
```

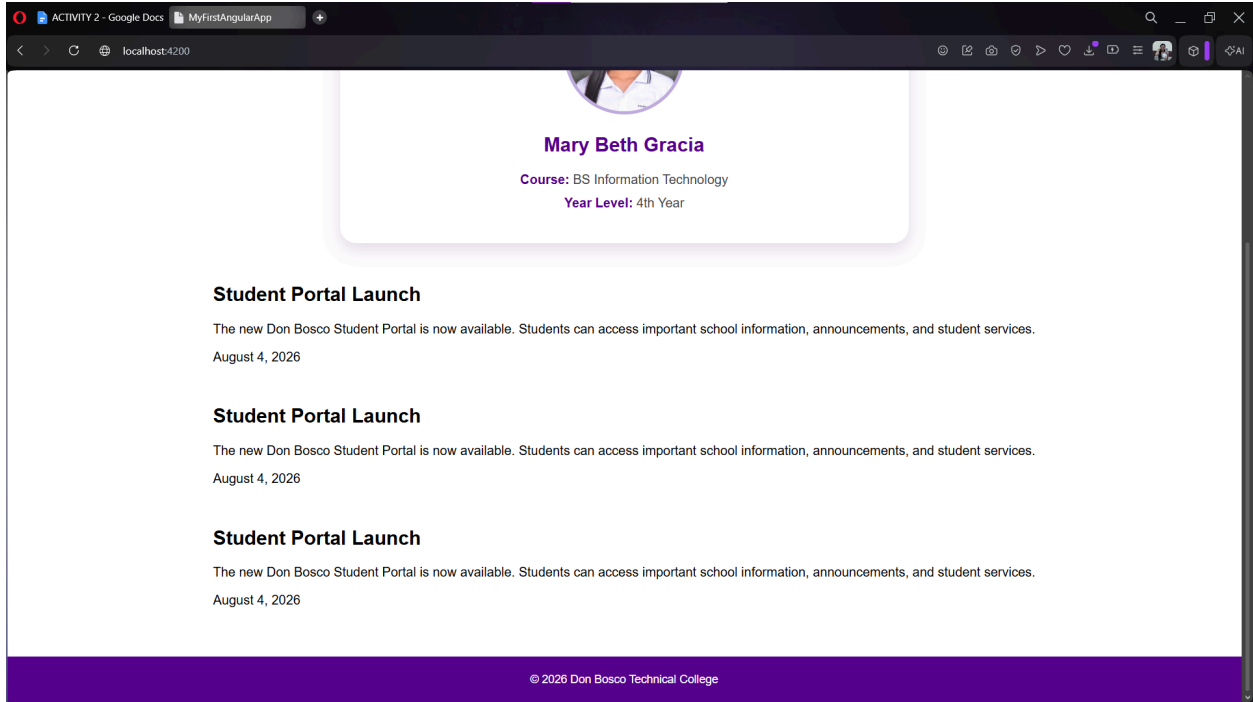
```
<app-announcement-card></app-announcement-card>
```



Evidence

- Browser
- App component showing three announcement cards





Flag 9 – Assemble the Dashboard

Your `app.html` should now act only as the page assembler.

Assemble your page to have this structure:

header

profile-card

announcement-card

announcement-card

announcement-card

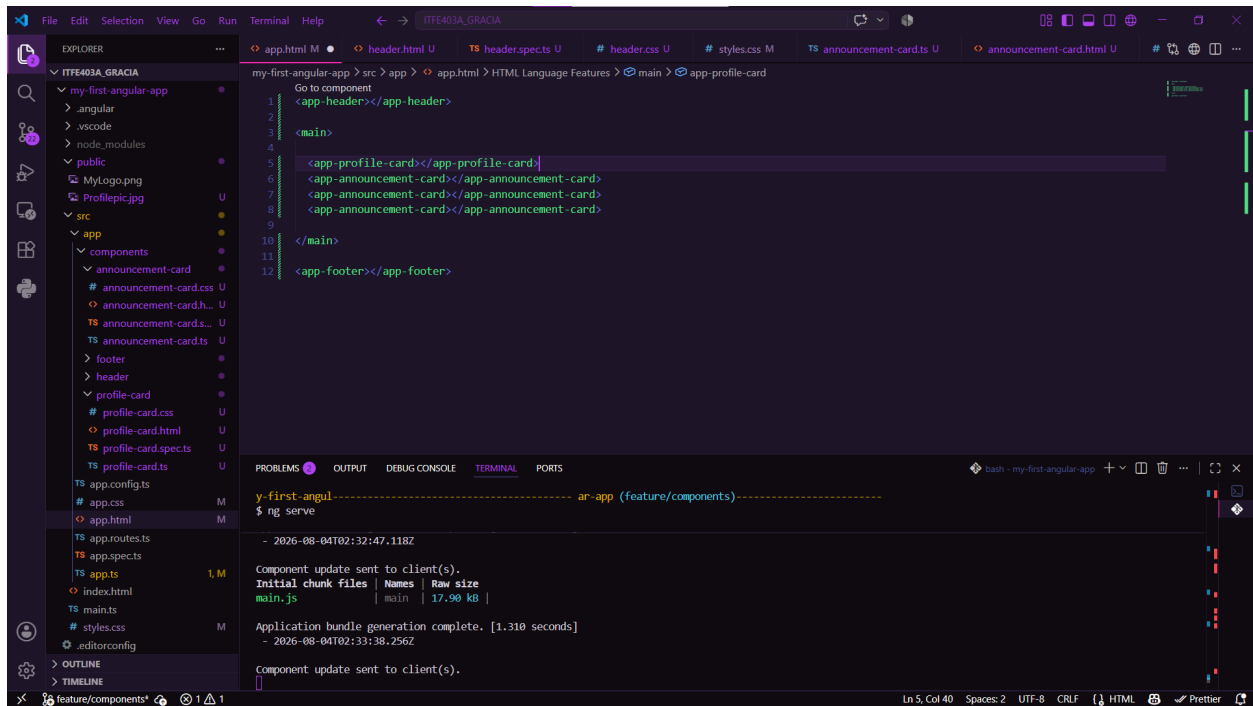
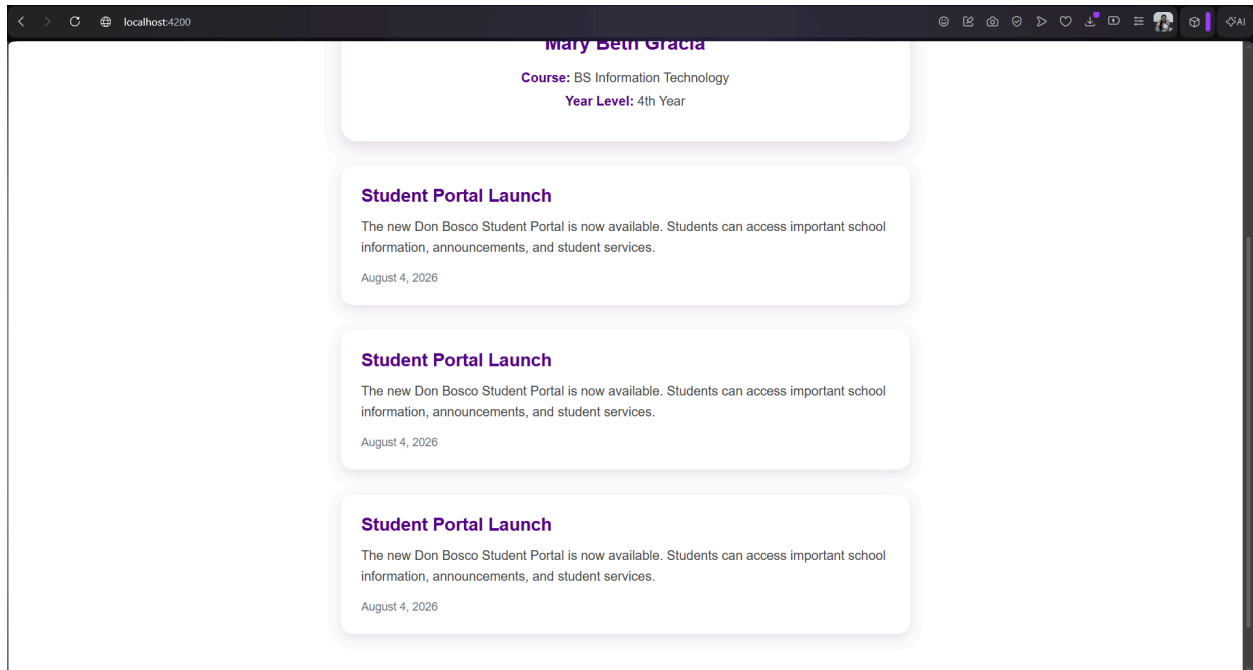
footer

Notice that almost no actual page content exists here. Only reusable components.



Evidence

- Browser
- `App.html`



Flag 10 – Client Changed Their Mind

Mission

The client sends an urgent request:

"Every announcement card should now have:

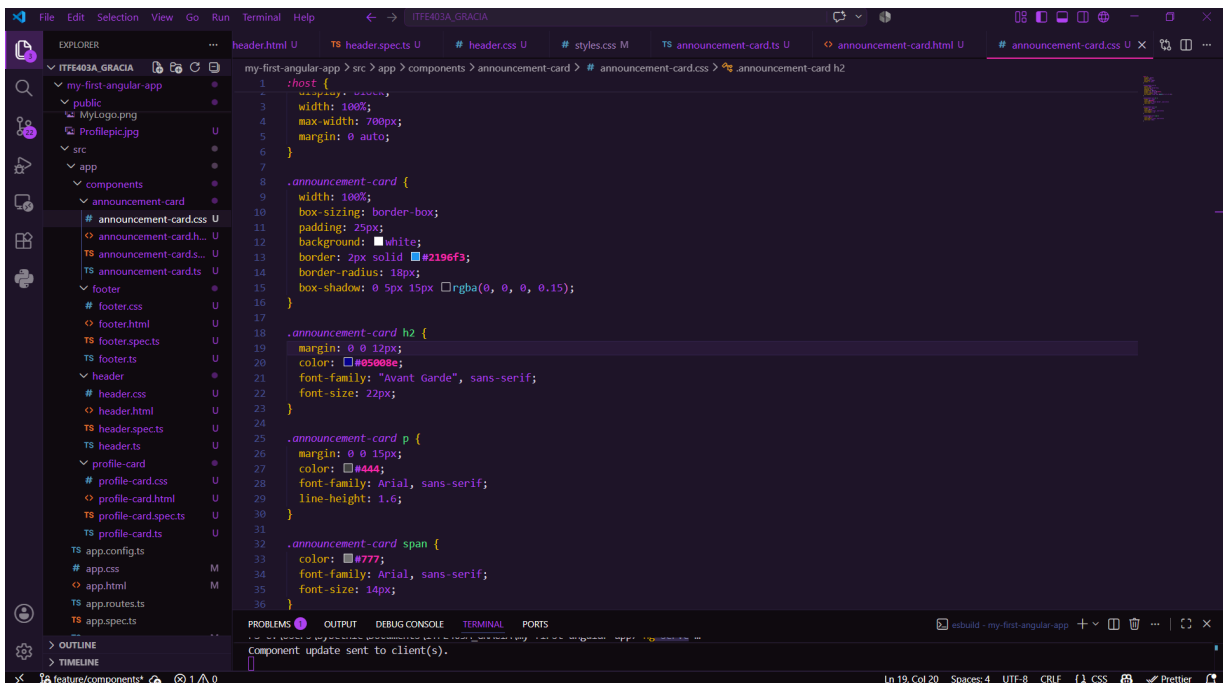
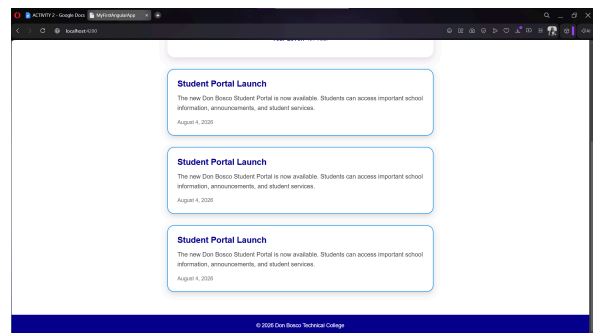
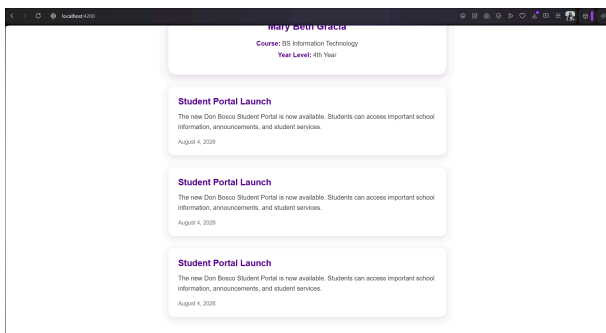
- Rounded corners
- A blue border
- Padding
- A subtle shadow"

You must update **only one file**.

Apply your styles there and verify that **all announcement cards** update automatically.

Evidence

- Browser before and after the CSS change
- **Announcement-card.css** code



Flag 11 – Pushing Changes

Commit your completed work, then push to your repository.

Evidence

- Terminal commands for committing + pushing
- Screenshot of the Github Repository reflecting the pushed changes

```
byethio@byethio: ~/Documents/ITFE403A_GRACIA/my-first-angular-app (feature/components)
$ git remote -v
origin https://github.com/beth7254/ITFE403A_GRACIA.git (fetch)
origin https://github.com/beth7254/ITFE403A_GRACIA.git (push)

byethio@byethio: ~/Documents/ITFE403A_GRACIA/my-first-angular-app (feature/components)
$ git log --online --decorate -5
fatal: unrecognized argument: --online

byethio@byethio: ~/Documents/ITFE403A_GRACIA/my-first-angular-app (feature/components)
$ git log --online --decorate -5
1111a0f HEAD -> feature/components feat: Activity 2 Angular

byethio@byethio: ~/Documents/ITFE403A_GRACIA/my-first-angular-app (feature/components)
$ git status -sb
## feature/components

byethio@byethio: ~/Documents/ITFE403A_GRACIA/my-first-angular-app (feature/components)
$ git push -u origin feature/components
Enumerating objects: 51, done.
Counting objects: 100% (51/51), done.
Delta compression using up to 8 threads
Compressing objects: 100% (51/51), done.
Writing objects: 100% (51/51), 465.45 KiB | 11.88 MiB/s, done.
Total 51 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
remote: Create a pull request for 'feature/components' on GitHub by visiting:
remote: https://github.com/beth7254/ITFE403A_GRACIA/pull/new/feature/components
remote:
to https://github.com/beth7254/ITFE403A_GRACIA.git
 * [new branch] feature/components -> feature/components
branch 'feature/components' set up to track 'origin/feature/components'.
```

```
byethio@byethio: ~/Documents/ITFE403A_GRACIA (feature/components)
$ git branch
bash: syntax error near unexpected token ';'

byethio@byethio: ~/Documents/ITFE403A_GRACIA (feature/components)
$ git branch
angular_activity
* feature/components
main

byethio@byethio: ~/Documents/ITFE403A_GRACIA (feature/components)
$ git remote -v
origin https://github.com/beth7254/ITFE403A_GRACIA.git (fetch)
origin https://github.com/beth7254/ITFE403A_GRACIA.git (push)

byethio@byethio: ~/Documents/ITFE403A_GRACIA (feature/components)
$ dir
FLAG_ACTIVITY.pdf ITFE403A_GRACIA.pdf my-first-angular-app

byethio@byethio: ~/Documents/ITFE403A_GRACIA (feature/components)
$ cd my-first-angular-app

byethio@byethio: ~/Documents/ITFE403A_GRACIA/my-first-angular-app (feature/components)
$ git branch
* feature/components
main

byethio@byethio: ~/Documents/ITFE403A_GRACIA/my-first-angular-app (feature/components)
$ git remote -v
origin https://github.com/beth7254/ITFE403A_GRACIA.git (fetch)
origin https://github.com/beth7254/ITFE403A_GRACIA.git (push)

byethio@byethio: ~/Documents/ITFE403A_GRACIA/my-first-angular-app (feature/components)
$ git add .
```

